DEPARTMENT OF WAR

JOINT ORDNANCE TEST PROCEDURE (JOTP)-057

GUIDELINES FOR SAFETY DESIGN CRITERIA FOR REMOTELY CONTROLLED FUZING SYSTEMS USED IN MUNITIONS

DOD Fuze Engineering Standardization Working Group (FESWG)

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JOTP-057
August 26, 2026

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26 August-2026	Joint Ordnance Test Procedure (JOTP)-057 Guidelines for Safety Design Criteria for Remotely Controlled Fuzing Systems used in Munitions

PREPARING ACTIVITY:	SPONSORING ACTIVITY:
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ABSTRACT:
This document provides guidance on additional safety architecture criteria for fuzes and safety and arming devices that are remotely controlled to permit capabilities for safe passage, recovery, and overhead safety operations, etc..

COORDINATION DRAFT REVIEWED BY:
This document was coordinated with the following Standardization Offices: AM, AR, AS, AV, AVN, CH, CR, CR6, DA, DH, EC, MC, MD1, MD3, MDA, MR, OS, PT, TE, AF-1, AF-10, AF-11, AF-18, AF-24, AF-27, AF-37, AF-40, AF-99, and AF-241. In addition, the document was also coordinated with the Joint Weapon Safety Working Group and select Subject Matter Experts (SMEs).

ASSIST COORDINATION DATE: 08 June-2026

IMPLEMENTATION PLAN:
1. This JOTP will be referenced in MIL-STD-1316, Fuze Design, Safety Criteria For and in MIL-STD-1911, Hand Emplaced Ordnance Design, Safety Criteria For.

2. This document has been developed by the DOD Fuze Engineering Standardization Working Group (FESWG) for use by the Army Fuze Safety Review Board (AFSRB), the Air Force Nonnuclear Munitions Safety Board (NNMSB), and the Navy Fuze and Initiation System Technical Review Panel (FISTRP) to evaluate fuzes and safety and arming devices employed in munitions that are remotely controlled.

APPROVING AUTHORITY:

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1. SCOPE

1.1. Scope. The purpose of this document is to establish additional design safety criteria for fuzes and safety and arming (S&A) devices employed in munitions that are remotely controlled to permit capabilities for safe passage, recovery, overhead safety operations, etc. This may include Remotely Controlled (RC) Fuze Systems requiring multiple arm-enabled and safe transitions during operation. The RC Fuze(s) performance during operational state changes shall provide safety throughout the platform's lifecycle environments associated with assembly, handling, storage, transportation, emplacement, post emplacement maintenance, and recovery. This document does not replace, but rather supplements, the fuze design safety criteria of MIL-STD-1316 and MIL-STD-1911. This document is in addition to the requirements of MIL-STD-1316 and MIL-STD-1911, as applicable.

1.2. Standardization Group. This document has been developed by the DoD Fuze Engineering Standardization Working Group (DoD FESWG) for use by the Army Fuze Safety Review Board (AFSRB), the Air Force Nonnuclear Munitions Safety Board (NNMSB), and the Navy Fuze and Initiation System Technical Review Panel (FISTRP).

1.3. Reviewing Authority. The Service Safety Authorities (SSA) will review fuze and S&A designs for acceptability using the guidance contained herein. Individual services may issue regulations or instructions which impose additional design safety criteria. Each program is encouraged to engage the appropriate SSA to determine the applicability of the guidance as well as any pertinent policies and/or additional requirements. Continued dialogue should occur with the SSA as the design matures to facilitate the safety approval / certification process.

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2. DEFINITIONS AND CLARIFICATION STATEMENTS

2.1. Definitions. For interpretation of this document, the following specific definitions apply.

2.1.1. RC Fuze System. The RC Fuze System consists of an RC Fuze, a communication link and an RC Device.

2.1.2. Arm-enabled. An RC Fuze is arm-enabled when the arming environments are validated, and the arming delay is satisfied in accordance with MIL-STD-1316 and MIL-STD-1911 and is ready to accept command(s) to arm.

2.1.3. Communication link. The communication path between the RC Device and the RC Fuze which may consist of multiple internal and/or external interfaces.

2.1.4. Hazardous transition. A change of state of the RC Fuze that increases the probability of arming and/or fuze functioning.

2.1.5. Operational state. A static snapshot of the RC Fuze (e.g., condition of the safety features, firing stimulus, etc.) that exists at any given time. The RC Fuze operational states are defined in 2.3.

2.1.6. Power. The following definitions apply for RC Fuze Power:

a. Power OFF. All fuze power is off.
b. Initialization Power. Power to safety-critical components, in accordance with SSA policy, is off.
c. Power ON. All fuze power is on.
d. Allowed Power. Fuze power, in accordance with SSA policy, is on.

2.1.7. Recovery. The return of the munition/platform for reuse or other disposition after a deployment.

2.1.8. RC Device. The device that commands and wirelessly or remotely controls the RC Fuze.

2.1.9. RC Fuze. A fuze (fuzing system), as defined in MIL-STD-1316, that performs multiple operational state changes controlled by the RC Device.

2.1.10. Non-Armed. An operational state of the RC Fuze with safety features capable of preventing unintentional arming, that includes:

a. Explosive train interruption. For explosive train interruption, the interrupter is in the full interrupted position in accordance with MIL-STD-1316.
b. Non-interrupted explosive train. For non-interrupted explosive train, there is no accumulated energy on the firing capacitor.

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2.1.11. Strong data typing. A fault tolerant technique wherein discrete or variable data is represented by a bit pattern that is unique for each valid value and cannot be confused with any other valid value even as a result of up to a three-bit error.

2.2. Clarification Statements. The RC Fuze performs multiple operational state changes controlled by the RC Device, building upon the safety tenets of MIL-STD-1316 and MIL-STD-1911, with the addition of remote command arming. RC Fuze state criteria and transitions are introduced to permit capabilities for allowable multiple arm-enable and safe transitions, safe passage, recovery, and overhead safety operations, etc. The following section (see 2.3) provides clarification on RC Fuze Operational States. Figure 1 contains a notional Munition/Platform Phase and RC Fuze Operational State diagram depicting both notional munition/platform phases and the associated RC Fuze operational states. The RC Fuze's operational states and state transitions should be defined using Figure 1 as reference.

2.3. RC Fuze Operational States.

2.3.1. FUZE OFF. The RC Fuze is unpowered, i.e., Power OFF (see 2.1.6).

2.3.2. FUZE INITIALIZED. If required, a state with Initialization Power (see 2.1.6) where mission data is applied.

2.3.3. FUZE UNARMED. A non-armed state with Allowed Power available (see 2.1.6) and all safety features in their original condition.

2.3.4. ARMING INITIATED. A non-armed state with Power ON (see 2.1.6), the safety features have commenced directly sensing environments and/or User actions and are ready to be validated and arming delay is ready to be achieved.

2.3.5. ARM-ENABLED. If required, a state where arming environments have been validated and arming delay has been achieved. The RC Fuze is ready to accept command(s) to arm. An RC Fuze employing explosive train interruption is considered arm-enabled when the safety features are satisfied; however, the explosive train remains interrupted. An RC Fuze employing non-interrupted explosive train is considered arm-enabled when at least one arming energy interrupter is enabled.

2.3.6. ARMED. Any firing stimulus can produce RC Fuze function.

2.3.7. ARMING DISABLED. If required, a non-armed state with Allowed Power (see 2.1.6), in which re-validation of prior arming environment(s) is required, if practical. Retention of prior arming environment criteria that cannot be re-validated is permitted.

2.3.8. FUZE DISARMED. If required, a non-armed state with Allowed Power (see 2.1.6), that includes the following:

a. All prior arming criteria / transition state data cleared.

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b. Single fault tolerant to entry into arm progression.
c. Communications link is available with restrictions during handling and manual operations to prohibit acting upon state transition commands.

2.3.9. RECOVERABLE. If required, a terminal non-armed state, with Power OFF, that includes the following:

a. Incapable of state transition until re-deployment.
b. Incapable of arm progression.
c. Reported RC Fuze Status Reporting (see 4.7) status sent prior to being powered off.
d. If required, manually operable safety features critical to fuze system safety are restored/reversed, (e.g., reasserting safety pin, etc.) prior to being powered off.

2.3.10. FAILED-SAFE. A terminal non-armed state, resulting from a safety-critical error, that is incapable of state transition. If possible, the RC Fuze has reported safety status.

2.3.11. STERILIZED. If required, a terminal non-armed state where the RC Fuze is incapable of state transition and has been rendered permanently incapable of functioning. If possible, the RC Fuze has reported safety feature status, or the munition/platform has reported the RC Fuze has been successfully rendered inoperable.

2.4. Munition/Platform Phases.

2.4.1. MISSION START. The RC System is prepared for intended use.

2.4.2. INITIALIZED. If required, initialization/configuration to enable mission data (e.g., communication link, GPS coordinates, ID numbers, etc.) setting prior to remote command and control is allowed.

2.4.3. DEPLOYED. The munition/platform is executing the mission.

2.4.4. PASSAGE / OPERATIONAL MAINTENANCE. If required, this phase addresses program requirements to provide for overhead safety, safe entry, and/or safe passage of personnel/materiel and for conduct of munition/platform maintenance activities to sustain operational readiness (e.g., replace batteries, adjust sensors, reinstall GPS location data, etc.).

2.4.5. RECOVERABLE. If required, the munition/platform(s) is deemed safe and allowed to be retrieved when remote command and control have ended and all munition/platform(s), RC Fuze (s) under RC Device control have been accounted for.

2.4.6. MISSION END. All payload(s) employed in the munition/platform have been expended and/or rendered inoperable.

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2.5. Notional Munition/Platform Phase and RC Fuze Operational State Diagram. Due to the various combinations of states, Figure 1 is intended as a representative diagram of a notional Operational State configuration. Figure 2 defines the symbol legend. The munition/platform phases correlate to the RC Fuze Operational States (depicted via the vertical dotted lines). For further description details of the diagram, see Appendix C.

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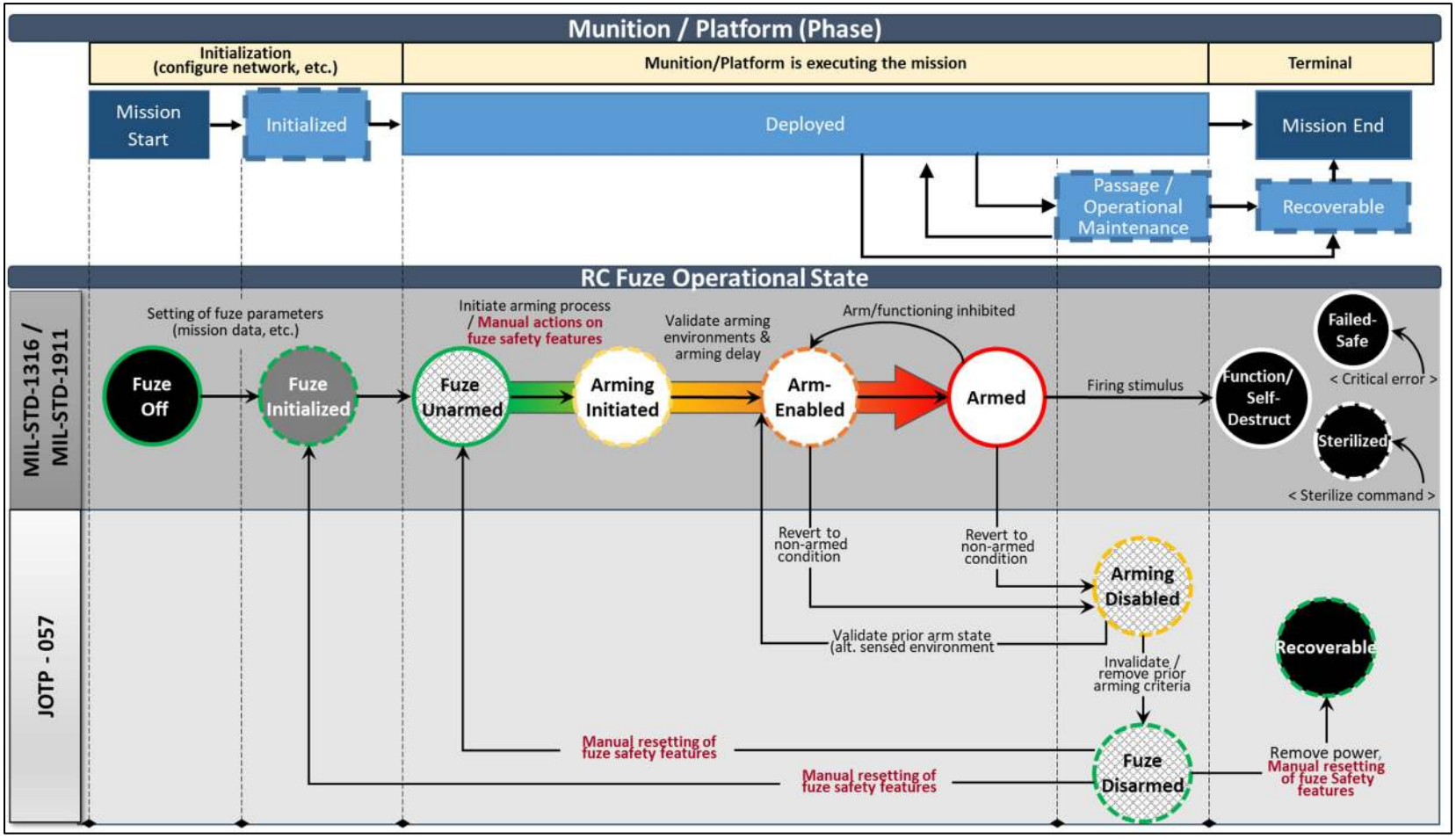


FIGURE 1. Notional Munition/Platform Phase and RC Fuze Operational State Diagram

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LEGEND						
Munition/Platform (Phase)		**RC Fuze Operational State**				
		RC Fuze State			**Text**	
	Nominal Phase		Solid perimeter line (green, red or white)	Nominal state		Arm progression
	Optional / System Specific Phase		Dashed perimeter line (green, yellow, orange, red or white)	Optional / System Specific state	abc	Fuze action
	Phase description		Black filled in circle	Power: OFF	**abc**	Unique actions by the User IAW MIL-STD-1911
			Gray filled in circle	Power: Initialization		
			While filled in circle	Power: ON		
			Cross hatched filled circle	Power: Allowed		

FIGURE 2. Legend for Notional Munition/Platform Phase and RC Fuze Operational State Diagram

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3. REFERENCED AND RELATED DOCUMENTS

3.1. REFERENCED DOCUMENTS

a. MIL-STD-1316, Fuze Design, Safety Criteria for
b. MIL-STD-1472, Design Criteria Standard for Human Engineering
c. MIL-STD-1911, Hand-Emplaced Ordnance Design, Safety Criteria for
d. DoD JOTP-051, Technical Manual for the use of Logic Devices in Safety Features
e. DoD JOTP-052, Guideline for Qualification of Fuzes, Safe and Arm (S&A) Devices, and Ignition Safety Devices (ISD)
f. DoD JOTP-054, Guidelines for the Design of Low Voltage Command Arm Distributed Fuzing Systems

3.2. RELATED DOCUMENTS

a. MIL-STD-331, Fuze and Fuze Components, Environmental and Performance Tests for (RELATED)
b. DoD JOTP-053, Electrical Stress Test

(Copies of these documents are publicly available online at https://assist.dla.mil)

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4. REQUIREMENTS

4.1. RC Fuze System Introduction. The RC Fuze System consists of an RC Fuze, a communication link and an RC Device. The RC Device controls the RC Fuze through the communication link. The RC Fuze is employed in a munition system which may be a stand-alone device, or which may be integrated onto a platform. Munition system configurations may have multiple RC Fuzes within one munition and/or have multiple munition systems on a platform. The platform (e.g., unmanned aerial vehicle (UAV), unmanned surface vehicles (USV), etc.) contains the munition system(s), power sources, sensors, and other components, as necessary. For reference, a diagram of the RC System is shown in Figure 3 below.

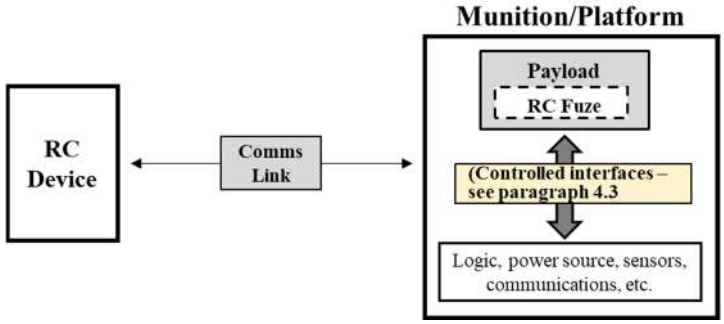


FIGURE 3. RC System

In view of the various possible combinations of an RC System, Figure 3 is intended as a basic representative RC System configuration. Other configurations shown in Appendix A are derivatives of Figure 3. The final design configuration shall be presented to the SSA for review and approval of compliance with this standard.

4.2. RC Fuze General Requirements.

4.2.1. An RC Fuze(s) design shall conform to the requirements of MIL-STD-1316, and/or MIL-STD-1911 as applicable.

4.2.2. Control and operation of the Fuze safety features, IAW MIL-STD-1316 and/or MIL-STD-1911, not validated and/or satisfied shall result in transition into the FAILED-SAFE state.

4.2.3. All required energy control features shall be co-located within the RC Fuze.

4.2.4. IAW MIL-STD-1316, the RC Fuze shall use independent and dissimilar logic circuitry to act upon any safety-critical signals (e.g., arm-enabled, status, etc.) to mitigate common cause and common mode failures.

4.2.5. Any safety-related logic implemented within the RC Fuze shall not be erasable, alterable or reconfigurable by credible environments that the RC Fuze would otherwise survive. Additional requirements for the implementation of safety-related logic are given in Joint

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Ordnance Test Procedure JOTP-051 (Technical Manual for the Use of Logic Devices in Safety Features).

4.2.6. An RC Fuze shall fail safe when it detects a safety-critical error.

4.2.7. RC Fuze shall prohibit arming or functioning while in non-armed states.

4.2.8. Use of probabilistic software and/or artificial intelligence, including machine learning is prohibited from controlling operation of the Fuze safety features, other than controlling an additional gate/condition for fuze arming (i.e., a final arming command issued by a mission computer.)

4.2.9. It is preferred that power to the RC Fuze be controlled and partitioned between safety-critical and non-safety-critical power in accordance with SSA policy.

4.2.10. An RC Fuze shall detect and reject (i.e., not act upon) all safety-critical messages that are corrupted (including a corrupted identifier), stale, or received out-of-sequence.

4.2.11. If the state transition signal and/or command is invalid, i.e., due to incorrect message or sequencing or a timeout occurs, the RC Fuze shall reject the state transition signal and/or command.

4.2.12. The RC Fuze shall not subvert the safety features (i.e., provide power to any safety switches prior to intended arming) as a result of status checks, operational maintenance procedures, or initialization sequence. In addition, stimuli shall not be provided to initiators except for the purpose of their initiation.

4.2.13. Anti-tamper features of the munition/platform or interfaces shall not degrade RC Fuze safety.

4.2.14. If the munition/platform contains multiple payload effector requiring the use of a distributed fuze system employing low voltage electrical signals to arm the fuze for subsequent initiation, the RC Fuze(s) design shall conform to the requirements of JOTP-054 (Guidelines for the Design of Low Voltage Command Arm Distributed Fuzing Systems).

4.3. RC Fuze Controlled Interface. RC Fuze controlled interface addresses the unique interface signals or commands/messages sent within the munition/platform, as represented in Figure 3. The following applies to these unique interface signals or commands/messages.

4.3.1. All RC Fuze safety-critical signals shall be robust such that the signal is not susceptible to inadvertent generation and subversion by credible environments anticipated in the lifecycle.

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4.3.2. There should be no inadvertent transmission of RC Fuze signals resulting from exposure to credible stressing electrical stimuli and electromagnetic environments.

4.3.3. All RC Fuze safety-critical messages shall be isolated from non-safety-critical messages in a robust configuration (i.e., physical separation of message communication lines). The method of implementation to show sufficient isolation between safety-critical messages and non-safety-critical messages shall be provided to the appropriate SSA for review and approval.

4.3.4. Upon unexpected loss of communication with the RC Device, the RC Fuze shall operate in accordance with the applicable system requirements.

4.3.5. The RC Fuze shall not be capable of acting upon manually operable safety features (e.g., manual switches, prior User actions that enable arming, etc.) when in the FUZE DISARMED (i.e., munition system is in PASSAGE / OPERATIONAL MAINTENANCE Phase) or RECOVERABLE state.

4.3.6. The RC Fuze shall not act upon state transition commands during performance of operational maintenance procedures.

4.4. RC Fuze State Criteria. The following lists the criteria required for FUZE UNARMED, ARMING DISABLED, FUZE DISARMED and RECOVERABLE states.

4.4.1. The FUZE UNARMED state criteria shall include the following:
- a. The RC FUZE is in the non-armed condition (see 2.1.10).
- b. Allowed Power (see 2.1.6) is applied.
- c. All safety features are in their original condition.

4.4.2. The ARMING DISABLED state criteria shall include the following:
- a. The RC FUZE is in the non-armed condition (see 2.1.10).
- b. Allowed Power (see 2.1.6) is applied.
- c. Prior arming criteria is retained when that arming environment cannot be re-validated.
- d. Transition into a more hazardous state is not suspectable to single point failure.

4.4.3. The FUZE DISARMED state criteria shall include the following:
- a. The RC FUZE is in the non-armed condition (see 2.1.10).
- b. Allowed Power (see 2.1.6) is applied.
- c. Prior arming criteria is invalidated/removed.
- d. Non-manual safety features are restored.

4.4.4. The RECOVERABLE state criteria shall include the following:
- a. The RC FUZE is in the non-armed condition (see 2.1.10).
- b. Fuze power is off, i.e., Power OFF (see 2.1.6).
- c. All arming criteria are invalidated/removed.

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d. All safety features, to include manual features, that have been previously removed and/or deactivated are restored/reversed.
e. The RC Fuze is incapable of receiving communications.

4.5. RC Fuze State Exit Transition Criteria.

4.5.1. Exiting from ARM-ENABLED or ARMED into the ARMING DISABLED state, the RC Fuze shall:
a. Revert to the non-armed condition (see 2.1.10).
b. For interrupted explosive trains, verify the interrupter is returned to the full interrupted position within the precision of detection.
c. For non-interrupted explosive trains, validate dissipation of the energy on the firing capacitor within the precision of the measurement.

4.5.2. Exiting from the ARMING DISABLED state into ARM-ENABLED, the RC Fuze shall:
a. Verify the presence of at least one unique post-deploy environment unless the environment(s) is unavailable. The rationale for use of environments and detection methods selected shall be provided to the SSA.
b. Validate prior state was either ARM-ENABLED or ARMED or validated arming criteria.
c. Permit this hazardous transition only as a consequence of intentional user actions.

4.5.3. Exiting from the ARMING DISABLED state into FUZE DISARMED, the RC Fuze shall:
a. Invalidate/remove prior arming criteria.
b. Restore non-manual safety features.
c. Permit and/or remove Allowed Power (see 2.1.6).

4.5.4. Exiting from FUZE DISARMED into FUZE INITIALIZED or FUZE UNARMED state, the RC Fuze shall:
a. Permit and/or remove Allowed Power (for FUZE UNARMED) or permit Initialization Power (for FUZE INITIALIZED) (see 2.1.6).
b. Verify the restoration/reversal of manually operable safety features critical to fuze system safety (e.g. reinsert safety pin, reattach shorting clips, etc.).

4.5.5. Exiting from FUZE DISARMED into RECOVERABLE state:
a. RC Fuze shall be powered off, i.e., Power OFF (see 2.1.6).
b. Manually operable safety features critical to fuze system safety, (e.g. reinsert safety pin, reattach shorting clips, remove batteries, etc.) shall be restored/reversed by the user.

4.5.6. Exiting from RECOVERABLE state is prohibited via remote command only.

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4.6. Safety Features Status Detection.

4.6.1. The RC Fuze shall provide means of detecting the status of the following aspects of the Safety Features, as applicable.

a. Energy control features or interrupter locks
b. Interrupter position
c. Arming power
d. Firing stimulus level

4.6.2. Detection methods shall:

a. Not compromise safety or degrade the energy control features or interrupter locks.
b. Not subvert the energy control features (i.e., non-isolated connections to the energy interrupters gates would not be acceptable).
c. Use the most direct measurement practical.
d. Not negate nor compromise the MIL-STD-1316 requirement for electrical firing energy dissipation.
e. Be executed in near real-time to ensure the reported status is not affected by stale data.

4.7. RC Fuze Status Reporting. Related to the requirements in 4.6, and 4.11, the following requirements establish the minimum design and evaluation criteria for reporting RC Fuze status indication.

4.7.1. The RC Fuze(s) shall include a means of reporting RC Fuze operational state and Safety Feature status (see 4.6).

4.7.2. Single point, common mode, or common cause faults shall not result in a false non-armed indication.

4.7.3. The RC Fuze shall report the state (e.g., ARMING DISABLED, ARM-ENABLED, ARMED, etc.), safety features status, and should include other related status information (e.g., unknown arm status, failure, function attempt, etc.)

4.7.4. The method of reporting status indication shall:

a. Not degrade safety.
b. Mitigate for use of stale data.

4.8. RC Fuze Safety System Failure Rate.

4.8.1. The RC Fuze safety system failure rates shall be in accordance with MIL-STD-1316 and/or MIL-STD-1911.

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4.8.2. For the ARMING DISABLED state, the RC Fuze safety system failure rate shall be as low as practical and consistent with the risk established as acceptable for unintended munition functioning.

4.8.3. For FUZE DISARMED or RECOVERABLE states, the RC Fuze safety system failure rate shall be less than one failure to prevent arming or functioning in one million fuzes.

4.9. Analysis and Qualification of RC Fuze Systems.

4.9.1. In addition to demonstrating compliance with MIL-STD-1316 and/or MIL-STD-1911, analyses shall be performed to identify hazardous conditions unique to RC Systems for the purpose of their elimination or control.

4.9.2. Compliance with this JOTP document shall be validated through conduct of tests and/or appropriate analyses in accordance with JOTP-052 (Guideline for Qualification of Fuzes, Safe and Arm (S&A) Devices, and Ignition Safety Devices (ISD)). Testing shall include the life cycle environments (i.e., arming delay through maintenance, recovery) and additional life cycle environments for any subsequent tactical employment reuse, if applicable. Qualification test plans shall be submitted to the appropriate SSA for approval.

4.10. RC System General Requirements. The RC System (see 4.1) is the aggregate of elements in a remotely controlled weapon system, that consists of the RC Fuze, munition/platform (as applicable), and RC Device. In addition to executing mission requirements, these elements also perform safety-critical functions to address hazards that may result from their inadvertent operation which could yield significant safety risk to personnel and costly materiel damage. Hazardous operation can be the result of inadvertent or premature fuze system arming, false safety status, and/or loss of accountability/control, that can occur in all configurations and conditions during operational use. For an example of the RC System safety-critical functions, see Appendix **B**. The following are general requirements and guidance criteria for the RC System and components within the scope of this document.

4.10.1. The RC System's User interface(s) shall be designed in accordance with MIL-STD-1472 and human factors engineering principles to minimize the risk of inadvertent hazardous activations. The interface(s) should be organized intuitively, clearly, and efficiently to support safe operation.

4.10.2. Software development shall be in accordance with accepted high quality software development procedures, such as AOP-52 or Joint Software System Safety Engineering Handbook.

4.10.3. The error rate of safety-critical messages received by the RC Fuze should be less than or equal to 1E-6.

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4.10.4. The RC System should include system health monitoring to ensure that safety-critical faults can be detected and reported to the operator.

4.10.5. The RC System should isolate safety-critical from non-safety-critical functions.

4.10.6. RC System safety-critical messages shall not be susceptible to inadvertent generation nor subversion due to the effects of credible environments anticipated in the lifecycle.

4.10.7. The RC System shall account and mitigate for any RC Fuze Status Reporting (see 4.7) latency.

4.10.8. The time requirements for safety-critical messages shall be defined in accordance with program requirements. Time synchronization errors should be minimized for safety-critical messages.

4.10.9. The RC System shall maintain identification of all RC Fuze(s) or munition/platform(s) throughout the mission.

4.10.10. The RC System shall utilize unique embedded electronic logic identifiers for all RC Fuze(s) or munition/platform(s). The following shall apply:

a. Unique identifiers shall utilize strong data typing.
b. Assigned unique identifiers shall not be capable of being altered or reassigned (while deployed) in any state other than FUZE INITIALIZATION state.

4.10.11. RC System safety system failure is defined as the failure of the RC System to prevent inadvertent transitions to the ARM-ENABLED or ARMED state. The RC System safety system failure rate shall be as low as practical or mitigated to an acceptable level of risk.

4.11. RC Fuze Status Indication. If the RC System design includes a RC Fuze safety status indicator residing on the RC Device and/or munition/platform, the following requirements apply.

4.11.1. The RC Fuze safety status indicator shall:

a. Provide a positive, unambiguous indication of the RC Fuze(s) status.
b. Provide the status information of all RC Fuzes (see 4.7) based upon applicability of operational requirements and/or User request.
c. Provide a safety status indicator of the RC Fuze (s) to include, but not limited to, the RC Fuze Status Reporting (see 4.7), RC Fuze unique identifier (see 4.10.10), and communication status.
d. Not display a safe indication unless all RC Fuze(s) are in a verified non-armed state. The probability of the RC Fuze status indicator falsely displaying the safe indication shall be as low as practical or mitigated to an acceptable level of risk.

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4.11.2. The RC Fuze safety status indicator should immediately notify the operator upon unexpected communications loss with the RC Fuze(s) and/or munition/platform.

4.11.3. When an RC Fuze status report indicates a failure, the RC Fuze safety status indicator should require User acknowledgment based on applicability of operational requirements.

4.11.4. If visual indication of the RC Fuze status resides on multiple components of the RC System (i.e., both the RC Device and munition/platform), User operating procedures should include a means to address conflicting RC Fuze status indications between the RC Device and munition/platform indicator.

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Appendix A
Notional RC System Configurations

Due to the various combinations of an RC System, all subsequent configurations shown below are derivatives of Figure 3. Figure A-1 below outlines an RC System in which more than one munition is under the control of the RC Device.

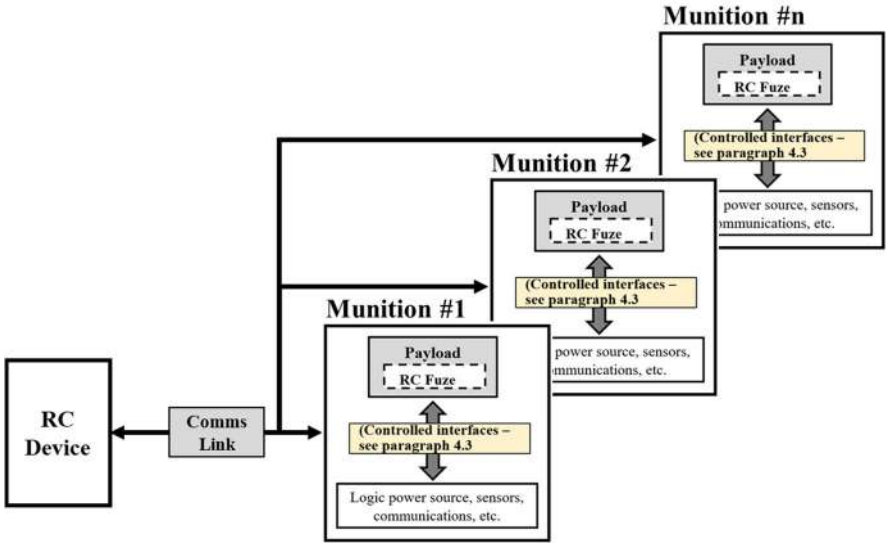


FIGURE A-1. RC System Configuration with Multiple Munitions

Figure A-2 below outlines an RC System when the platform contains multiple payloads. Similar to figure A-1, there may be more than one platform under the control of the RC Device.

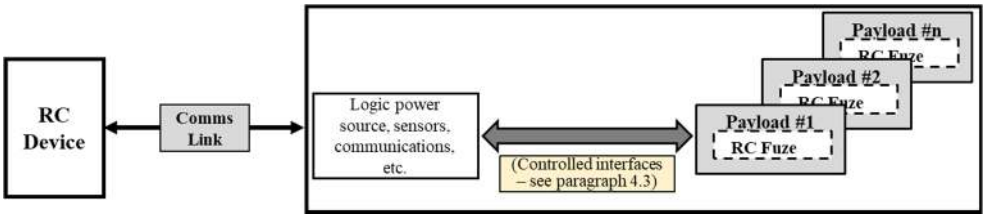


FIGURE A-2. RC System Configuration with Multiple Payloads

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Appendix B
Safety-Critical Functions of a Notional RC System

Table 1 provides an example of safety critical functions that may be performed by elements of a notional RC System. To derive the safety-critical functions, a preliminary hazard analysis should be performed on the RC system and its unique program specifications to identify and classify hazards posed by credible life-cycle environments, as well as conditions and personnel actions that may occur. Due to the varied combinations of an RC System, all possible configurations should be considered. For example, RC Systems that perform grouping (e.g., swarm, field, etc.) that utilize autonomous behavior, i.e., information sharing or decisions among the RC System components that are outside the User's authority, to commence hazardous state transitions should be addressed.

SAFETY-CRITICAL FUNCTIONS OF A NOTIONAL RC SYSTEM		
Hazard	**RC Fuze**	**RC System**
Inadvertent ARM	a. Reject invalid state transition b. Reject inadvertent/corrupt arm commands c. Transition to or maintain a safe state when faults occur. d. Validate arming environments and state criteria	a. Prevent invalid control of arming b. Preclude generation of erroneous, inadvertent, or corrupted commands c. Inhibit activation commands issued by non-authoritative sources d. Detect inadvertent or incorrect asset identification
Inadvertent launch / dispense	NA	a. Prevent invalid control of launch/dispense b. Preclude generation of erroneous, inadvertent, or corrupted commands c. Detect inadvertent or incorrect asset identification
Inadvertent autonomous lethality engagement	NA. RC Fuze should be in ARM-ENABLED or ARMED state	a. Prevent inadvertent activation of autonomous modes b. Detect inadvertent or incorrect asset identification c. Detect false reporting of self-destruct time remaining
System does not accept User commands	NA	a. Preclude generation of erroneous, inadvertent, or corrupted commands
Control by more than one User at a given time	NA	a. Preclude generation of erroneous, inadvertent, or corrupted commands b. Inhibit activation commands issued by non-authoritative sources c. Preclude inadvertent transfer of control from one User to another without operator authorization
False display of status	a. Detect accurate RC Fuze Status b. Prevent report of invalid/corrupt message of RC Fuze Status	a. Prevent erroneous display of RC Fuze Status b. Detect inadvertent or incorrect asset identification c. Maintain direct and positive control of munitions.

TABLE 1. Safety-Critical Functions of a Notional RC System

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Appendix C
Notional Munition/Platform Phase and RC Fuze Operational State Diagram Description

The Notional Munition/Platform Phase and RC Fuze Operational State Diagram (see Figure 1) is intended to illustrate the allowable RC Fuze operational states and state entrance and exit criteria required for RC Fuze non-armed states. The diagram was developed to represent the minimum execution of RC Fuze non-armed states allowed by the SSA. The following provides a detailed explanation of how to read and interpret the diagram.

The Notional Munition/Platform Phase and RC Fuze Operational State Diagram consist of two horizontal sections: Munition/Platform Phase and the RC Fuze Operational States which is subdivided into MIL-STD-1316/MIL-STD-1911 fuze states and JOTP-057 fuze states. Vertical dotted lines (used as swim lanes) bound segments of the two horizontal sections to correlate the associated Munition/Platform Phase relative to the RC Fuze Operational States. The two horizontal sections are explained as follows:

a) The Munition/Platform Phase (top horizontal section) depicts a munition/platform tactical deployment from mission start to mission end. Each phase is represented by a blue colored rectangle; a solid perimeter of the rectangle indicates a nominal phase of the system, whereas a dashed perimeter represents an optional or system-specific phase. The munition/platform phases include:
   i. Mission Start and Initialized: These phases are intended to enable system configuring/initializing of the munition/platform to the RC Device and set mission data (e.g., GPS coordinates, ID numbers, etc.) prior to tactical remote command and control.
   ii. Deployed: This phase represents the munition/platform's tactical emplacement and operation.
   iii. Passage / Operational Maintenance: An optional or system-specific phase, which may facilitate overhead safety and/or entry or safe passage of friendlies. For the munition/platform, maintenance activity to sustain operational readiness may be allowed to include handling and manual operations required to sustain critical mission needs (e.g., battery change, repositioning, etc.) performed after manually operable safety features critical to fuze system safety are reversed, (e.g. reasserting safety pin, etc.).
   iv. Mission End and Recoverable: These phases represent the end of the munition/platform's operational deployment and tactical use.

b) The RC Fuze Operational State (middle horizontal section) is subdivided to illustrate and differentiate between the MIL-STD-1316/MIL-STD-1911 fuze states and JOTP-057 fuze states, explained as follows:
   i. The MIL-STD-1316 /MIL-STD-1911 fuze states depict the arm progression succession. Each fuze state is represented by a circle. The circle's color perimeter represents the hazard level, i.e., less hazardous (green) to most hazardous (red); a solid perimeter indicates a nominal state, whereas a dashed perimeter represents an optional or system-specific state. The circle's color fill

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represents power categories as explained in the legend. The arrows represent the exit and entrance criteria among states. System-specific states that are not needed based on system requirements can be considered pass-throughs; however, all nominal state transition criteria requirements still apply (see 4.4 and 4.5) when passing through unused states. (NOTE: Entrance into FAILED-SAFE or Sterilized state is permitted at any time.) For MIL-STD-1911 fuze systems that require User actions to manually set or reset fuze safety features (e.g., turning of a switch, safety pin insertion, etc.), those actions are represented by red text accompanied by arrows.

ii. The JOTP-057 fuze states depict the unique states allowed in accordance with this document. Similar to the color-coding scheme used for the MIL-STD-1316 / MIL-STD-1911 fuze states, the circle's color perimeter represents the hazard level, i.e., less hazardous (green) to more hazardous (yellow). All optional or system-specific states are represented by the dashed perimeter. These optional or system-specific states may not be needed based on system requirements. However, if such states exit as pass-throughs, then the state transition requirements (see 4.4 and 4.5) still apply.

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